Enrollment nnumber - 240840116004

practical 3 : Apply filtering,sorting , aggregation and groupby operatons on real world dataset using pandas

```
import pandas as pd
data = { "Emp_id" : [101,102,103,104,105,106,107,108],
         "name" : ["rahul","priya","amit","riya","karan","neha","vivek","pooja"],
         "department" : ["IT","HR","IT","Finance","HR","Finance","IT","HR"],
         "city" : ["Surat","Navsari","Surat","Valsad","Surat","Navsari","Valsad","Surat"],
         "salary" :[45000,38000,52000,48000,40000,51000,60000,42000],
         "Expereince" : [2,1,5,4,2,6,8,3]
         }
df = pd.DataFrame(data)
print(df)
```

```
   Emp_id   name department     city  salary  Expereince
0     101  rahul         IT    Surat   45000           2
1     102  priya         HR  Navsari   38000           1
2     103   amit         IT    Surat   52000           5
3     104   riya    Finance   Valsad   48000           4
4     105  karan         HR    Surat   40000           2
5     106   neha    Finance  Navsari   51000           6
6     107  vivek         IT   Valsad   60000           8
7     108  pooja         HR    Surat   42000           3
```

Double-click (or enter) to edit

step 1

```
df
```

Show hidden output

step 2 : display employees from it departement

```
df[df["department"]=="IT"]
```

Show hidden output

```
df[df["Emp_id"]==103]
```

Show hidden output

display employee whoise salary is greater than 50000

```
df[df['salary']>50000]
```

Show hidden output

display emp-loyees from surat

```
df[df['city']=="surat"]
```

	Emp_id	name	department	city	salary	Expereince
0	101	rahul	IT	surat	45000	2
2	103	amit	IT	surat	52000	5
4	105	karan	HR	surat	40000	2

APPLY MULTIPLE CONDITIONS : display employees from the it department , having salary greater than 50000

```
df[(df["department"]=="IT")&(df["salary"]>50000)]
```

	Emp_id	name	department	city	salary	Expereince
2	103	amit	IT	Surat	52000	5
6	107	vivek	IT	Valsad	60000	8

sort salary in assecnding order

```
df.sort_values(by="salary", ascending=True)
```

	Emp_id	name	department	city	salary	Expereince
1	102	priya	HR	Navsari	38000	1
4	105	karan	HR	Surat	40000	2
7	108	pooja	HR	Surat	42000	3
0	101	rahul	IT	Surat	45000	2
3	104	riya	Finance	Valsad	48000	4
5	106	neha	Finance	Navsari	51000	6
2	103	amit	IT	Surat	52000	5
6	107	vivek	IT	Valsad	60000	8

sort salary in descending order

```
df.sort_values(by="salary",ascending=False)
```

	Emp_id	name	department	city	salary	Expereince
6	107	vivek	IT	Valsad	60000	8
2	103	amit	IT	surat	52000	5
5	106	neha	finance	Navsari	51000	6
3	104	riya	Finance	Valsad	48000	4
0	101	rahul	IT	surat	45000	2
7	108	pooja	HR	Surat	42000	3
4	105	karan	HR	surat	40000	2
1	102	priya	HR	Navsari	38000	1

sort by deparetmenmt and salary

```
df.sort_values(by=["department","salary"] , ascending=[True,False])
```

	Emp_id	name	department	city	salary	Expereince
5	106	neha	Finance	Navsari	51000	6
3	104	riya	Finance	Valsad	48000	4
7	108	pooja	HR	Surat	42000	3
4	105	karan	HR	surat	40000	2
1	102	priya	HR	Navsari	38000	1
6	107	vivek	IT	Valsad	60000	8
2	103	amit	IT	surat	52000	5
0	101	rahul	IT	surat	45000	2

step 4 : perform aggregation

```
calculate the total salary of all the employee
```

```
df["salary"].sum()
```

```
np.int64(376000)
```

find the highest salary

```
df["salary"].max()
```

```
60000
```

find lowest salary

```
df["salary"].min()
```

```
38000
```

count employees

```
df["Emp_id"].count()
```

```
np.int64(8)
```

step 5 - perform groupby operations

```
Start coding or generate with AI.
```

Average salary department-wise

group employees based on department

calculate the average salary of each department

```
df.groupby("department")["salary"].mean()
```

	salary
department	
Finance	49500.000000
HR	40000.000000
IT	52333.333333

dtype: float64

min salary department wise

```
df.groupby("department")["salary"].min()
```

department	salary
Finance	48000
HR	38000
IT	45000

dtype: int64

count employees department wise

```
df.groupby("department")["Emp_id"].count()
```

department	Emp_id
Finance	2
HR	3
IT	3

dtype: int64

max salary

```
df.groupby("department")["salary"].max()
```

department	salary
Finance	51000
HR	42000
IT	60000

dtype: int64

total salary department wise

```
df.groupby("department")["salary"].sum()
```

	salary
department	
Finance	99000
HR	120000
IT	157000

dtype: int64

average salary city wise

```
df.groupby("city")["salary"].mean()
```

	salary
city	
Navsari	44500.0
Surat	44750.0
Valsad	54000.0

dtype: float64

step 6 : Multiple agrregation functions

```
df.groupby("department")["salary"].agg(["count", "sum", "min", "max", "mean"])
```

	count	sum	min	max	mean
department					
Finance	2	99000	48000	51000	49500.000000
HR	3	120000	38000	42000	40000.000000
IT	3	157000	45000	60000	52333.333333

step 7 - save the result

```
df.to_csv("Emp_iddata.csv" , index=False)
print("dataset saved succesfully ")
```

```
dataset saved succesfully
```

step 8 dowload csv file

```
df.to_csv("Emp_Id.csv",index=False)
from google.colab import files
```

```
files.download("Emp_Id.csv")
```